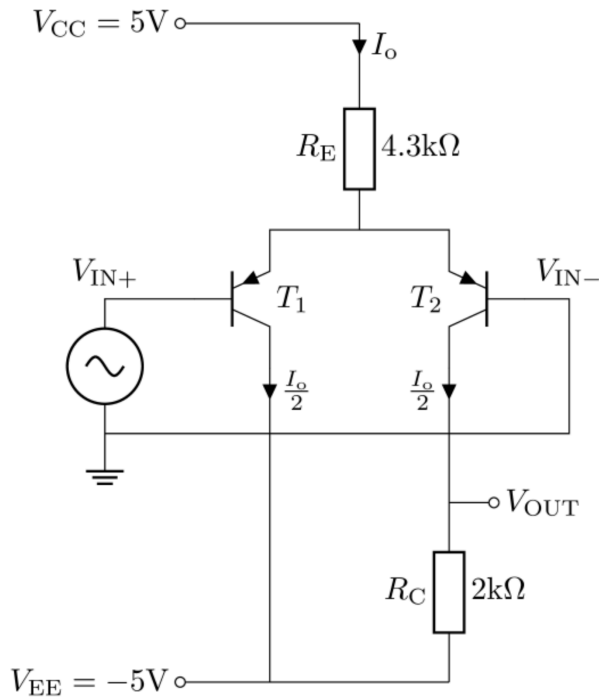


Section 3: Differential Amplifier

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Introduction

An op-amp input stage must amplify the difference between two inputs (differential mode) while rejecting signals common to both (common mode). The long-tailed pair (LTP) achieves this: with equal inputs, IC splits equally between T_1 and T_2 ; a differential input unbalances the currents, producing a large change in V_{out} . A common-mode signal also changes V_{out} , but the large emitter resistor R_E provides degeneration that suppresses it. Circuit: T_1 (V_{in+}) and T_2 (V_{in-}) are 2N2907 PNP transistors with shared emitter current I_o through $R_E = 4.3\text{ k}\Omega$. V_{out} is taken at the collector of T_2 across $R_C = 2\text{ k}\Omega$. Supply: $\pm 5\text{ V}$.



Differential Gain

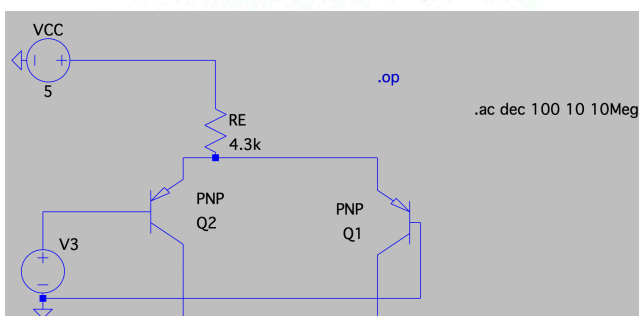
Theoretical Analysis:

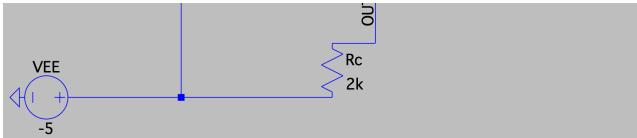
From the small-signal model, the differential voltage gain is:

$$A_d = \frac{v_o}{v_d} = \frac{g_{m,T2}}{2} \left(\frac{1}{r_{o,T2}} + \frac{1}{R_C} \right)^{-1}$$

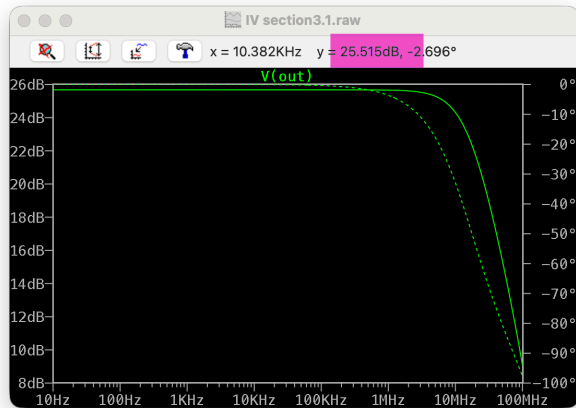
$$A_d = \frac{v_o}{v_d} = \frac{g_{m,T2}}{2} \left(\frac{1}{r_{o,T2}} + \frac{1}{R_C} \right)^{-1}$$
$$\text{As } I_{C,T2} = \frac{I_o}{2} = 0.5 \times 10^{-3} \text{ A.}$$
$$g_m = \frac{I_C}{V_T} = \frac{0.5 \times 10^{-3}}{0.025} = 0.02 \text{ S.}$$
$$r_o = \frac{V_A}{I_C} = \frac{120}{0.5 \times 10^{-3}} = 240 \text{ k}\Omega.$$

$$\therefore A_d = 0.01 \times \left(\frac{1}{240 \text{ k}} + \frac{1}{2000} \right)^{-1} = 19.83.$$
$$\text{And } 20 \log_{10}(19.83) = 25.9 \text{ dB.}$$





The LTspice AC simulation gave a measured differential gain of 25.515 dB, in close agreement with the theoretical value of 25.95 dB.



Differential Input Resistance

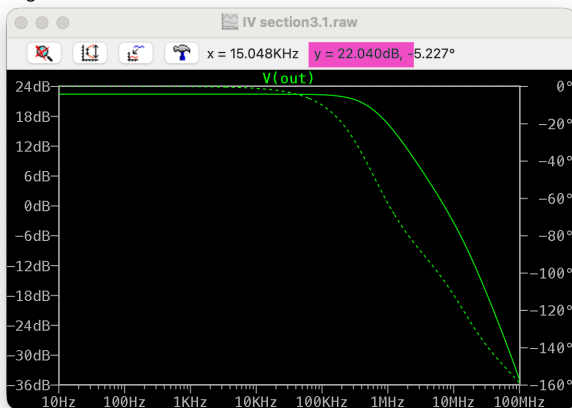
From the small-signal model:

$$r_{ind} = 2r_{be} = \frac{2\beta}{g_m}$$

Rind = 25kohm

$r_{ind} = 2r_{be} = 2\beta / g = 2 \times 250 / 0.02 = 25 \text{ k}\Omega$

A 10 kΩ resistor was added in series with the base of T₁, reducing the gain to 22.040 dB. Using the gain-change method:



$$A'_V = A_V \frac{r_{in}}{r_{in} + 10k}$$

$A_V = 25.473\text{dB} = 18.77$
 $A'_V = 22.040\text{dB} = 12.65$

$$r_{in} = 10k \frac{A'_V}{A_V - A'_V}$$

So $r_{in} = 20.67\text{kohm}$, can be considered as matching the theoretical prediction of 25 kΩ.

Common-Mode Gain and CMRR

Common-Mode Gain

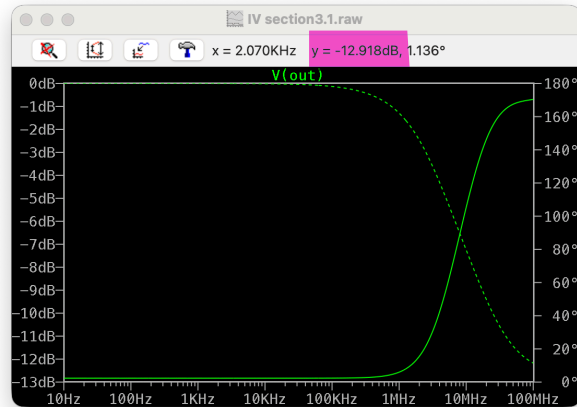
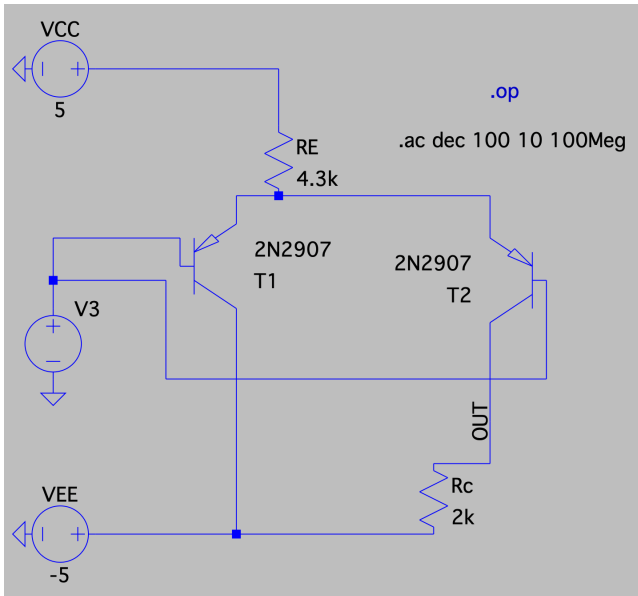
The small-signal model gives the common-mode gain:

$$A_c = \frac{v_o}{v_c} = -\frac{R_C}{2R_E}$$

$$A_c = v_o / v_c = -R_C / (2R_E)$$

$$A_c = -2k / (2 \times 4.3k) = -0.233 \Rightarrow -12.67 \text{ dB}$$

The LTspice AC simulation confirmed a common-mode gain of -12.918 dB .



Common Mode Rejection Ratio (CMRR)

The CMRR is the ratio of differential gain to common-mode gain:

$$\text{CMRR} = \frac{A_d}{A_c} = g_m R_E = \frac{R_E I_o}{2V_T}$$

$$\text{CMRR} = A_d / A_c = g \cdot R_E = R_E \cdot I_o / (2V_T) = 0.02 \times 4.3k = 85.297 \Rightarrow 38.62 \text{ dB}$$

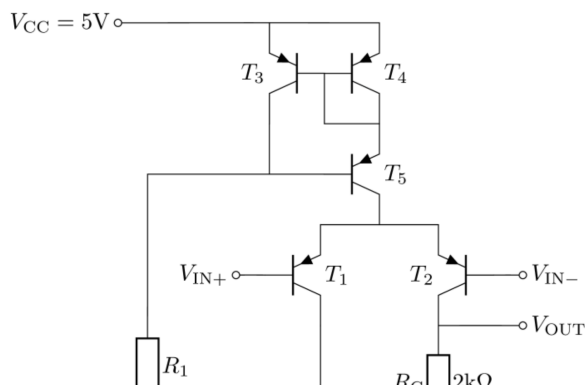
From the simulated gains:

$$\text{CMRR} = 10^{(25.675/20)} / 10^{(-12.8/20)} \approx 259 \Rightarrow 48.3 \text{ dB (measured values)}$$

Increasing CMRR and gain with active load

Circuit Modification

The tail resistor R_E was replaced by a 2N2907 current mirror (T_3 – T_5) supplying $I_o = 1 \text{ mA}$, with $R_1 = 8600 \Omega$. Replacing R_E with a high-impedance current source greatly increases the common-mode rejection ratio, because the current source maintains I_o nearly constant regardless of common-mode input, suppressing the common-mode gain.





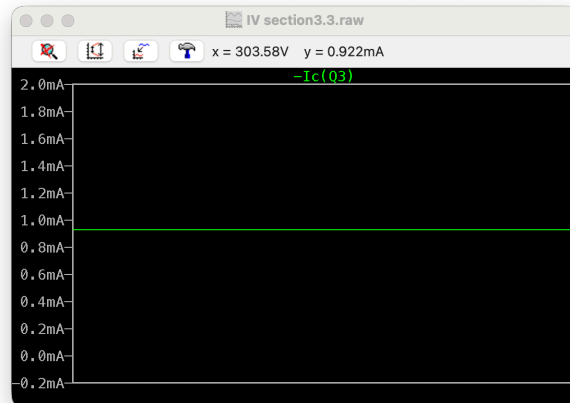
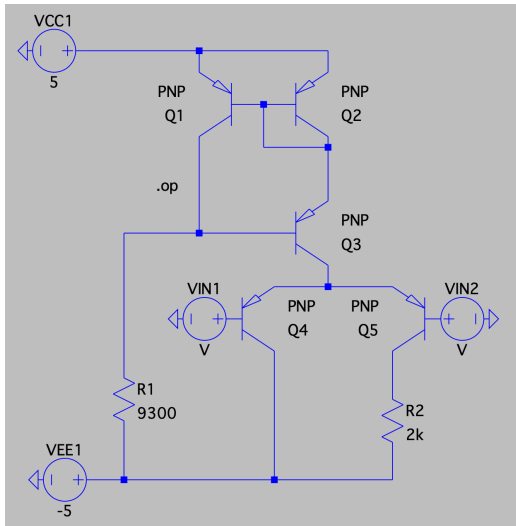
LTP with Active Tail — Results

- Differential gain: $A_d = 24.756\text{dB} = 17.290$
- Common-mode gain: $A_c = -92.633\text{dB} = 2.3353 \times 10^{-5}$

$$\text{CMRR} = A_d / A_c = 10^{(24.756/20)} / 10^{(-92.633/20)} = 740372.0 \Rightarrow 117.4 \text{ dB}$$

$$\text{So CMRR} = 740372.0 = 117.4 \text{ dB}$$

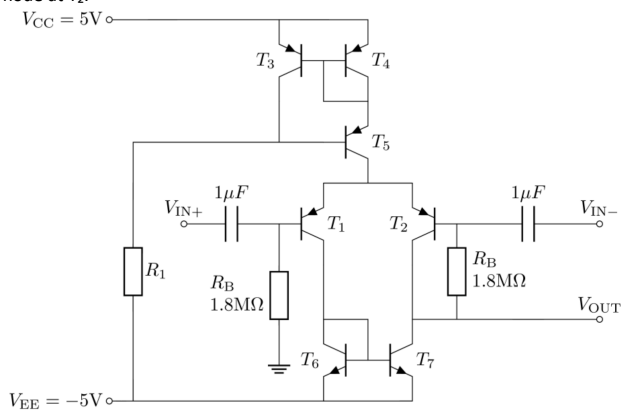
Replacing the tail resistor with a current mirror increased the CMRR from $\sim 48.3 \text{ dB}$ to 117.4 dB , a dramatic improvement. This occurs because the Wilson mirror has a very high output impedance, effectively acting as an ideal current source and eliminating the path through which common-mode signals would otherwise cause emitter voltage variations.



LTP with Second Current Mirror (Active Load on Collector)

Circuit and Analysis

A second current mirror (T_3 – T_4 – T_5 , PNP) was added as an active collector load, reflecting the collector current of T_1 into the output node at T_2 .



This doubles the effective transconductance seen at the output, significantly increasing the differential gain. The small-signal differential gain becomes:

$$A_d = g_{m,T2} \left(\frac{1}{r_{o,T2}} + \frac{1}{r_{o,T7}} + \frac{1}{R_B} \right)^{-1}$$

$$A_d = g_{m,T2} \cdot (1 / r_{o,T2} + 1 / r_{o,T7} + 1 / R_B)^{-1}$$

$$\text{With } g_{m,T2} = 1.077 \text{ A/V, } r_{o,T2} = 1.000 \text{ M}\Omega, r_{o,T7} = 1.000 \text{ M}\Omega, R_B = 1.8 \text{ M}\Omega$$

With $I_{C1} = I_{C2} = I_{E1} = 0.5 \text{ mA}$, $V_{A(\text{NPN})} = 100 \text{ V}$, $V_{A(\text{PNP})} = 200 \text{ V}$, $R_B = 1.8 \text{ M}\Omega$:

$$\begin{aligned} A_d &= (0.5 \text{ m} / 0.025) \cdot (0.5 \text{ m} / 100 + 0.5 \text{ m} / 200 + 1 / 1.8 \text{ M})^{-1} \\ &= 20 \cdot (5 \times 10^{-6} + 2.5 \times 10^{-6} + 5.56 \times 10^{-7})^{-1} \\ &\approx 2037.1 \Rightarrow 66.3 \text{ dB} \end{aligned}$$

The LTspice AC simulation confirmed a measured differential gain of 65.590 dB, in close agreement with the theoretical prediction of 66.3 dB.

Adding the second current mirror as an active collector load increased the midband differential gain from 25.95 dB (simple resistor load) to 65.590 dB — an improvement of over 40 dB — by exploiting the high output impedance of the current mirror to maximise the voltage developed at the output node.

